



FLEET & SHORE PERSONNEL TRAINING

Dangerous, Hazardous & Harmful Cargoes Training

The IMDG Code's classification, packaging, stowage and documentation rules, and how the ship prepares for and responds to an incident involving dangerous goods.

IMDG Code

SOLAS Ch. VII

MARPOL Annex III

Company SMS

What We Will Cover

01

IMDG Code Overview

Legal basis, structure and how the Code is used onboard

02

The Nine Classes of Dangerous Goods

Classification and the primary hazard each class presents

03

Packaging, Marking & Labelling

UN packaging, hazard labels, marks and placards

04

Stowage & Segregation

Stowage categories and the segregation table

05

Documentation

The DG manifest, container packing certificate and safety data sheets

06

Emergency Response — EmS & MFAG

Using the Emergency Response and Medical First Aid Guides



SECTION 1

IMDG Code Overview

01

Why every ship carrying packaged dangerous goods operates under one common international code.

Where the IMDG Code Comes From

The International Maritime Dangerous Goods Code is made mandatory by SOLAS and MARPOL, giving it binding legal force worldwide.

- SOLAS Chapter VII, Part A requires the carriage of dangerous goods in packaged form to comply with the IMDG Code.
- MARPOL Annex III sets requirements for preventing pollution by harmful substances carried in packaged form, closely aligned with the IMDG Code's marine pollutant provisions.
- The IMDG Code is amended every two years by the IMO to reflect new substances, packaging technology and lessons learned from incidents.
- It applies to all SOLAS ships carrying packaged dangerous goods, and is supplemented by the Company's own dangerous goods procedures in the SMS.
- Non-compliance can result in port denial, cargo refusal, PSC detention and, in the event of an incident, serious safety and pollution consequences.



Structure of the Code

The Code is organised in two volumes: Volume 1 covers general provisions, training, packing and stowage; Volume 2 is the Dangerous Goods List with individual UN number entries.

KEY DEFINITIONS

Terms Used Throughout This Training

Vocabulary that appears on every shipping document, label and stowage plan for dangerous goods.



UN Number

A four-digit number identifying a specific dangerous substance or a group of substances, e.g. UN 1203 (petrol).



Proper Shipping Name (PSN)

The official name used to describe the substance on transport documents — never a trade name or brand name.



Marine Pollutant

A substance identified as harmful to the marine environment, requiring the marine pollutant mark in addition to any class label.



Competent Authority

The national body empowered to approve packaging, issue exemptions and enforce dangerous goods regulations.



SECTION 2

The Nine Classes of Dangerous Goods

02

Every dangerous substance falls into one of nine classes, each with its own hazard characteristics and handling rules.

CLASSIFICATION

Classes 1 to 5

Each class label is a distinct colour and symbol so hazards are recognisable at a glance.

Class	Description	Example
1	Explosives	Fireworks, distress flares, ammunition
2	Gases (compressed, liquefied or dissolved)	Acetylene, LPG, oxygen cylinders
3	Flammable liquids	Petrol, paint, diesel additives
4	Flammable solids; substances liable to spontaneous combustion; substances emitting flammable gas when wet	Matches, calcium carbide, sodium
5	Oxidising substances and organic peroxides	Ammonium nitrate, hydrogen peroxide

CLASSIFICATION

Classes 6 to 9

The remaining classes cover toxicity, radioactivity, corrosivity and other miscellaneous hazards.

Class	Description	Example
6	Toxic and infectious substances	Pesticides, clinical/medical waste
7	Radioactive material	Radioactive isotopes, medical sources
8	Corrosive substances	Sulphuric acid, sodium hydroxide
9	Miscellaneous dangerous substances and articles (incl. marine pollutants)	Lithium batteries, dry ice, asbestos

SUBDIVISIONS

Classes With Divisions

Some classes are further divided by degree or type of hazard — the division number is part of the class marking.



Class 1 — Explosives

Divided 1.1 to 1.6 by hazard type, from mass explosion risk (1.1) to insensitive articles (1.6).



Class 2 — Gases

Divided 2.1 flammable, 2.2 non-flammable/non-toxic, 2.3 toxic gas.



Class 4 — Flammable Solids

Divided 4.1 flammable solids, 4.2 spontaneously combustible, 4.3 dangerous when wet.



Class 6 — Toxic Substances

Divided 6.1 toxic substances, 6.2 infectious substances.



SECTION 3

Packaging, Marking & Labelling

How a package tells everyone who handles it exactly what hazard is inside, without a word needing to be read.

03

UN Packaging Standards

Only packaging tested and certified to the applicable UN performance standard may be used for dangerous goods.

- Packaging is assigned a Packing Group (I, II or III) reflecting the degree of danger — I is highest, III is lowest — which determines the level of packaging performance required.
- Approved packaging carries a UN packaging code stamped or printed on it, confirming the type, packing group and test standard it meets.
- Damaged, leaking, corroded or otherwise defective packaging must never be loaded — it is rejected and reported before it reaches the ship.
- Limited quantity and excepted quantity provisions allow reduced marking/documentation for small, low-risk quantities of some substances.
- Overpacks combining several packages must still show all the marks and labels of the packages inside, unless those are visible through the overpack.



Read Before You Touch

Every crew member handling cargo should be able to recognise a UN number, hazard label and marine pollutant mark before opening or moving a package — not after.

Marks, Labels and Placards

Three distinct visual elements appear on dangerous goods packages and containers, each with its own purpose.



Marks

Text and symbols such as the UN number, proper shipping name and orientation arrows — informational, not hazard warnings.



Labels

Diamond-shaped hazard labels on individual packages showing the primary and any subsidiary hazard class.



Placards

Larger versions of hazard labels displayed on cargo transport units (containers, tank containers) carrying dangerous goods.



Marine Pollutant Mark

A separate black-and-white fish-and-tree symbol required whenever the substance is also classified as a marine pollutant.



SECTION 4

Stowage & Segregation

04

Positioning dangerous goods aboard so that incompatible substances are always kept safely apart.

Where Dangerous Goods Can Be Stowed

The IMDG Code assigns each substance a stowage category that determines where on the ship it may be carried.

- Stowage categories A to E define whether goods may be stowed on or under deck, and how far from accommodation, machinery spaces or the ship's side they must be kept.
- Some substances require mechanical ventilation, protection from heat sources, or a specified maximum stack height in the stowage plan.
- Stowage must also account for proximity to living quarters, escape routes and lifesaving appliances — dangerous goods can never block these.
- The ship's approved cargo securing manual and the IMDG Code's stowage requirements together determine final container or package placement.
- A pre-stowage plan is checked against the segregation table and stowage requirements before loading begins, not adjusted improvised on the quay.



Never Improvise Stowage

A late change to the stowage plan must be re-checked against segregation rules before it is accepted — a single overlooked incompatibility can be catastrophic.

THE SEGREGATION TABLE

Keeping Incompatible Classes Apart

The IMDG Code's segregation table sets the minimum separation required between classes that could react dangerously together.

Segregation Term	Meaning
Away from	Effectively separated so incompatible substances cannot interact in a casualty
Separated from	In different holds/compartments or on deck at a defined horizontal distance
Separated by a complete compartment or hold from	A full compartment or hold must separate the two substances
Separated longitudinally by an intervening complete compartment or hold from	The strictest requirement — used for the most dangerous combinations

Applying Segregation Correctly

Segregation is a planning discipline that continues right through loading and the voyage.



Do

- Check the segregation table for every class combination before finalising the stowage plan
- Verify actual container/package positions against the plan once loading is complete
- Keep dangerous goods documentation readily available to the bridge team throughout the voyage
- Reassess segregation whenever cargo is restowed, discharged or shifted mid-voyage



Don't

- Don't accept a late-loaded container without checking its segregation against what is already stowed
- Don't stow incompatible classes in the same hold based on convenience or lack of space
- Don't stow dangerous goods where they block emergency escape routes or firefighting access
- Don't rely on memory — always consult the current segregation table and stowage plan



SECTION 5

Documentation

The paperwork that must accompany every dangerous goods shipment, and why it must be accurate before the cargo is ever loaded.

05

What Must Accompany the Cargo

Dangerous goods documentation must be complete, accurate and available before loading is authorised.

- The Dangerous Goods Declaration (DGD) states the UN number, proper shipping name, class, packing group and quantity for each substance shipped.
- A Container Packing Certificate confirms the container was packed correctly, segregated internally and is properly marked and placarded.
- The ship's Dangerous Goods Manifest lists every dangerous goods shipment aboard, including its stowage position, and must be kept on the bridge.
- A Safety Data Sheet (SDS) for each substance provides hazard information, handling precautions and emergency measures for crew reference.
- Documentation is checked by the ship's officers before acceptance — cargo without complete, correct paperwork must be refused.



Manifest on the Bridge

A current, accurate Dangerous Goods Manifest must always be accessible on the bridge — it is the single most important reference in any dangerous goods emergency.

DOCUMENT CHECKLIST

Core Documents at a Glance

Four documents form the backbone of dangerous goods paperwork on any voyage.

DGD

Dangerous Goods Declaration —
shipper's statement of what is
being shipped and its classification

CPC

Container Packing Certificate —
confirms correct packing and
segregation inside the container

SDS

Safety Data Sheet — hazard,
handling and emergency
information for each substance

DGM

Dangerous Goods Manifest — the
ship's master list of everything
dangerous aboard



SECTION 6

Emergency Response — EmS & MFAG

06

Two ready-reference guides that tell the bridge team exactly what to do in the first minutes of a dangerous goods incident.

THE TWO GUIDES

EmS Guide and MFAG

Every substance in the Dangerous Goods List carries an EmS code and a reference to the medical guide — both must be consulted together.



EmS Guide

The Emergency Response Procedures for Ships Carrying Dangerous Goods gives fire (F) and spillage (S) schedules for each substance's UN number.



MFAG

The Medical First Aid Guide for Use in Accidents Involving Dangerous Goods gives first-aid procedures matched to each substance's medical first aid guide number.



F-Schedule (Fire)

Tells the crew whether to fight a fire involving the substance, what extinguishing agent to use, and any special precautions.



S-Schedule (Spillage)

Covers PPE, containment and clean-up actions for a leak or spillage of the substance.

On Discovery of a Dangerous Goods Incident

The first few minutes shape how a dangerous goods incident is contained.



Do

- Raise the alarm immediately and identify the substance from the manifest and stowage plan
- Consult the correct EmS F- and S-schedule before approaching the affected cargo
- Don appropriate PPE and use breathing apparatus where fumes or vapour are present
- Inform the Master, DPA and, where relevant, the nearest coastal authority without delay



Don't

- Don't approach a leaking or burning package without first identifying the substance and hazard
- Don't apply water or an extinguishing agent not specified by the substance's F-schedule
- Don't assume a small spillage is minor — check the S-schedule for reactivity and toxicity
- Don't delay reporting while attempting to resolve the incident alone

PREPAREDNESS

Being Ready Before an Incident Happens

The best response to a dangerous goods emergency is one the crew has already rehearsed.

- Officers responsible for cargo operations must be familiar with the EmS Guide and MFAG layout before a voyage carrying dangerous goods begins.
- The ship's emergency response equipment — spill kits, chemical-resistant PPE, breathing apparatus — is checked and stocked in line with the carried cargo.
- Musters and drills periodically include a dangerous goods incident scenario, not only fire and abandon-ship drills.
- Contact details for the Company DPA, shore chemical experts and the nearest coastal state authority are kept current and accessible on the bridge.
- Lessons from any near miss or actual dangerous goods incident are reported and fed back into the Company's SMS and crew training.



COURSE COMPLETE

Correct classification, packaging and stowage are what stand between a routine cargo operation and a serious casualty.

Company DPA — 24/7 emergency line

Nearest RCC / MRCC — via VHF Ch.16 / DSC

Fleet Safety Department — mng@gloryship.com.tr